# AM50 COMPUTER LAB 8

## 1. Introduction

The goal of today's computer lab is as follows:

*Introductory* We will read in data for the Standard and Poor's 500 Index into MATLAB, and ask whether there is evidence for or against the hypothesis that the data is well described by a random walk. We will see that the percent changes in the stock prices obey a random walk better than the changes in the stock price themselves

*Advanced* More elaborate discussion of whether Stock prices obey random walks, including a simple effort to estimate the diffusion constant (volatility) of the Standard and Poors index.

## 2. *Introductory* Fun with Stock Data: Exponential growth!

Let us start to study data from Stocks, and determine whether a random walker model has any validity. To read in data from the *S&P* 500 for 50 years use the command

```
M=csvread('S_P_500.csv',1,1);
```

This reads the data from January 3 1953 to February 19,2010. The data is saved for every day that the Stock Market is open. The first column $M(:,1)$ gives the opening price; the second column gives the high $M(:,2)$; the third column gives the low and the fourth $M(:,4)$ gives the close. The fifth column gives the volume of shares traded.

> *Note on how MATLAB stores matrices*: Matlab stores *everything* in matrices ("Mat" in MATLAB stands for "matrix", not "math"). Dimensions are read as (rows,columns). Thus, M(3,1) indicates the element in the 3rd row, 1st column of M. The colon : means everything. So M(:,1) indicates all rows, 1st column of M, so it is the first column of M.

For some strange reason the data is stored backwards–the early dates are at the end of the table and the end dates are at the beginning. Thus if you plot

```
plot(M(:,1),'o')
```

you will see that the data backwards–with the last date occurring first and the first date occurring last. To correct this lets type

```
plot(M(end:-1:1,1),'o')
```

This reverses the plot.

Now, the data looks to be quite exponential! This makes sense of course, you are usually told to invest in stocks (or even a bank) because of exponential growth. And even inflation is exponential. A simple first bit of analysis to do is to see how well the data is fit by the law

$$S(t) = S_0 e^{rt},$$

where $S(t)$ is the price of the stock portfolio at time $t$, $S_0$ is the initial price and $r$ is the growth rate.

Recall from earlier, the way to test that something is an exponential we should take the logarithm of the equation: We have

$$log(S(t)) = log(S_o e^{rt}) = log(S_0) + rt.$$

This if we plot $log(S)$ versus $t$, the data should fall on a straight line.

**Exercise: 1**
To test the idea that stock prices grow exponentially, lets make a logarithmic plot of the stock data. Type *semilogy(M(end:-1:1,1))*. You will see that the data is basically a straight line. Demonstrate that the data is pretty well fit by the law $S(t) = 20e^{rt}$, with $r = 0.0003$. According to this law, how much should Harvard's $\sim 20$ billion dollar endowment increase in size during a decade?

For the last demonstration, you need to plot $S(t) = 20e^{rt}$ on top of the measurements. To do this consider the following commands

```
r=3e-4;
t=0:100:15000; % declares a 'time variable"
s=20*exp(r*t);  % forms the exponential
semilogy(t,s);
```

*Matlab Remark:* Once you've entered these commands into MATLAB, type "t" and you will see that the time variable is an array of numbers between 0 and 15000; Correspondingly s is the value of the exponential function at these times.

## 3. Do Stock Prices Obey Random Walks

We now turn to discuss whether stock prices obey random walks. Recall what we stated in class during the random walk discussion: there are two principal properties of a random walks:

(1) Any variable that obeys a random walk should have a width obeying $L(t) = \sqrt{Dt}$
(2) The probability distribution for moving a distance $r$ after time $t$ is given by a normal distribution (or Gaussian)

$$p(r,t) \sim \frac{1}{\sqrt{Dt}} \exp\{-\frac{r^2}{4Dt}\}. \tag{1}$$

In both of these $D$ is the diffusion constant.

Let's see whether Stock's show any trace of this behavior. One obvious difference is what we discovered already in the last section: That Stock prices increase exponentially in time. *But the position of random walkers do not increase exponentially in time!* Thus we already have a bit of a problem with our model. On the other hand, maybe it is possible that Stock prices obey random walks anyways–and the growth is over longer times.

Let's consider and compare and contrast two possibilities:

(1) Stock prices obey random walks. If this is the case then we expect both of the laws mentioned above to apply to the stock prices themselves.

(2) Maybe it is not the stock prices themselves that obey random walks, but instead the rate at which stock prices grow. If we consider the exponential growth equation

$$\frac{dS}{dt} = r(t)S,$$

then this is tantamount to saying that *the percentage change* in stock price changes according to a random walk from moment to moment.

To test these possibilities, we first need to form a vector that contains the *jumps* in both the stock prices, and the percentage jumps in the stock prices over time. We can then figure out if each of them shows evidence of obeying a random walks. We write the code:

```
stocks=M(:,1);  % this is a vector of the stock prices
diff_stocks=diff(stocks);  % this makes a vector of the day-day
                                                 %  jumps in the stock prices
percent_diff_stocks=diff_stocks./(0.5*(stocks(1:end-1)+stocks(2:end)));
```

The variable

`diff_stocks`

contains the jumps in stock prices from day-day. The variable

`percent_diff_stocks`

contains the percentage jump in the stock prices.

Let's now make a histogram of these jumps and see if either is well approximated by the Gaussian in equation (1).

To do this, we first need to form a histogram: This can be done with the following commands

```
[hist_stocks,x_stocks]=hist(diff_stocks,[-10:0.01:10]);
[hist_stocks_percent,x_stocks_percent]=hist(percent_diff_stocks,[-1:0.001:1]);
```

*Remarks about the commands:* The command "hist" makes a histogram. The way we have set it up, both

`x_stocks`

and

`hist_stocks`

are arrays of numbers.

`hist_stocks(N)`

tells the number of times the jump in the stock price was in the interval

`[x_stocks(N),x_stocks(N+1)].`

We have chosen our intervals to be between $-10 \leq S \leq 10$ for stock prices, and within $\pm 1\%$ for percent changes. These bounds were invented to make the plots as clear as possible

**Exercise: 2**

Plot the histograms for both the changes in the stock prices and the percent changes.

For this question, you might use the command

```
plot(x_stocks,hist_stocks,'.')
```

How do you determine if they are well fit by Gaussian's? Here is a simple method: Suppose we have a Gaussian

$$p(S,t) = P_0 \exp\{-\frac{S^2}{4Dt}\}.$$

If you take the logarithm of both sides we have

$$log(p(S,t)) = log(P_0) - \frac{S^2}{4Dt}.$$

Thus the data will look like an upside down parabola!! Let's check this:

**Exercise 3**

Make logarithmic plots of the histograms for both the changes in the stock prices and the percent changes.Are they well fit by Gaussian's?

For this question, you might use the command

```
semilogy(x_stocks,hist_stocks,'.')
```

You should notice the following things in your data:

(1) The percent change data data looks exactly like an upside down parabola.
(2) The stock change data looks like a parabola when the price changes are small– above this though the data veers away from it significantly.

This is why the percent change model is preferred in our analysis of option pricing.

## 4. Advanced: More elaborate Checks of the Gaussian Model

OK-how can we check our model further? Thus far we have shown that the day-day stock price percent changes are well described by a Gaussian. To see if the day to day changes are well approximated by a random walk, we also need to check the *time dependence* in equation (1). Namely we need to check that the *width of the distribution increases in time according to* $\sqrt{Dt}$. *And what is the diffusion constant?*

### *Exercise 4*

*Redo the plots we have made above but instead of using 1 day intervals of stock prices, use 1 week and 1 month stock intervals (ie 5 and 20 days). Determine if the width is increasing in a manner consistent with our random walks. Can you estimate D?*

*Note that to do this you will need to plot the change in prices every 5 or 20 days. The easiest way to form an array with these differences is probably to redo the entire exercise using the following command as the first one:*

```
stocks_5day=M(1:5:end,1);
```

*This takes every fifth stock price, so every fifth day. If you repeat all of the exercises using this array instead of the stock array we made above, you will create a histogram of stock prices that can be compared against the ones generated in the first two sections.*